Computational PDEs — Day 3 homework

Exercises (about 45–60 minutes)

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Format

Two questions, both by hand.
Throughout, $0 < \rho < 1$ and $\zeta \neq 0$. The infinite product below may be reordered: for $\rho < |\zeta| < 1/\rho$ it converges absolutely.

1. **An identity for P** — inversion in the unit circle.
2. **Möbius maps** — why a Schottky generator is holomorphic, and why the cross ratio is unchanged.

1 Question 1 — Identities for $P(\zeta, \rho)$

The function used on the annulus is

$$P(\zeta, \rho) = (1 - \zeta) \prod_{n=1}^{\infty} (1 - \rho^{2n}\zeta)\,(1 - \rho^{2n}\zeta^{-1}). \tag{1}$$

(a) Inversion

Show that

$$P(\zeta^{-1}, \rho) = -\zeta^{-1}\, P(\zeta, \rho). \tag{2}$$

1. Write out $P(\zeta^{-1}, \rho)$. The infinite product contains the same factors as $P(\zeta, \rho)$, in another order.
2. Check the prefactor:

$$1 - \zeta^{-1} = -\zeta^{-1}(1 - \zeta). \tag{3}$$

3. Conclude the identity.

Q1(a) — the prefactor and one sentence for the product:

What this identity is for

On $|\zeta| = 1$ one has $\overline{\zeta} = 1/\zeta$, so (a) is inversion in the outer circle. Thursday's loss penalises this inversion identity at sample points.

2 Question 2 — Möbius maps and the cross ratio

A Möbius transformation is

$$\varphi(\zeta) = \frac{a\zeta + b}{c\zeta + d}, \qquad ad - bc \neq 0. \tag{4}$$

It is holomorphic away from the pole $\zeta = -d/c$.

On a disk with M circular holes, inversion in one circle alone uses a complex conjugate, so it is not holomorphic. The maps that generate the reflections are compositions of *two* inversions. Those compositions are Möbius.

(a) A Schottky generator is Möbius

For a hole of centre δ_m and radius $q_m > 0$,

$$\theta_m(\zeta) = \delta_m + \frac{q_m^2 \zeta}{1 - \overline{\delta_m}\,\zeta}. \tag{5}$$

1. Clear the denominator and write $\theta_m = (a\zeta + b)/(c\zeta + d)$. Do not rescale.
2. Compute $ad - bc$, and check that it equals $q_m^2 \neq 0$.

$a =$ _____, $b =$ _____, $c =$ _____, $d =$ _____

$ad - bc =$ _____

(b) The cross ratio is invariant

For four distinct points, the cross ratio is

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)/(z_1 - z_4)}{(z_2 - z_3)/(z_2 - z_4)}. \tag{6}$$

1. Prove the difference formula

$$\varphi(z) - \varphi(w) = \frac{(ad - bc)\,(z - w)}{(cz + d)(cw + d)}. \tag{7}$$

2. Substitute $z, w \in \{z_1, z_2, z_3, z_4\}$ into the cross ratio of the image points. The factors $ad - bc$ cancel, and so does every term $cz_j + d$. Conclude

$$\big(\varphi(z_1), \varphi(z_2); \varphi(z_3), \varphi(z_4)\big) = (z_1, z_2; z_3, z_4). \tag{8}$$

Q2(b) — the difference $\varphi(z) - \varphi(w)$, then the cancellation:

Why this is the multiply connected step

Four points lie on a circle or a straight line precisely when their cross ratio is real. Part (b) says a Möbius map preserves that, so it sends circles and lines to circles and lines. Each θ_m is such a map: it identifies the reflected copy of hole m with the hole itself, and it stays holomorphic. The prime function is the product over compositions of these maps.

Time box

About 45–60 minutes. If time is short, finish Q1(a) and Q2(b).